Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT **Paper -1(Adv-2022-P1-Model)** **Date: 21-12-2025**

Time: 09.00Am to 12.00Pm **GTA-08** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**4**	2	**4**	3	**0**	4	**5**	5	**5**	6	**3**
7	**3**	8	**6**	9	**AD**	10	**AB**	11	**AC**	12	**AC**
13	**ABC**	14	**ABC**	15	**B**	16	**A**	17	**D**	18	**C**

PHYSICS

19	**2.00**	20	**4.00**	21	**2.00**	22	**0.78-0.79**	23	**1.00-1.05**	24	**8.00**
25	**3.00**	26	**6.25**	27	**ABCD**	28	**CD**	29	**BCD**	30	**ABD**
31	**CD**	32	**ACD**	33	**B**	34	**C**	35	**A**	36	**A**

CHEMISTRY

37	**6**	38	**8**	39	**5**	40	**5**	41	**4**	42	**7.8 – 7.9**
43	**7**	44	**1.73 – 1.90**	45	**ABCD**	46	**AD**	47	**ACD**	48	**AB**
49	**ABC**	50	**AD**	51	**B**	52	**A**	53	**A**	54	**B**

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHEMATICS

1. Let $f=\pi\left\{\left(1+\sin^2\alpha\right)\left(1+\cos^2\alpha\right)\right\}$

Each of $\left(1+\sin^2\alpha\right)\left(1+\cos^2\alpha\right)$ has to have maximum value

i.e., $1+1+\frac{1}{4}\sin^2 2x\le\frac{9}{4}$ attains max at $\sin^2 2x=1$

$\Rightarrow x=\frac{\pi}{4},\frac{3\pi}{4}$ $\quad\therefore f_{\max}=\left(\frac{9}{4}\right)^4=m$ ___(1)

No of quadruples $(\alpha,\beta,\gamma,\delta)$ is $2\times2\times2\times2=2^4=16=n$ _____(2)

$\therefore G.E=\left[\sqrt[4]{\left(\frac{9}{4}\right)^4.2^4}\right]=[4.5]=4$

2. $\frac{dy}{dx}=\frac{4y^3+2yx^2}{3xy^2+x^3}$ $put\ y=ux\Rightarrow x\frac{du}{dx}=\frac{u^4+u}{3u^2+1}$

$\int\frac{3u^2+1}{u^3+u}du\Rightarrow\frac{1}{x}dx\Rightarrow u^3+u=cx\Rightarrow c=2\Rightarrow y^3+x^2y=2x^4$

Now $x=3\Rightarrow y^3+9y-16z=0\Rightarrow y\in[4,5)$ $\quad\left(\because f(4)<0;f(5)>0\right)$

$y(3)=?$ $\quad\therefore n=4$

3. $g(f(x))=x$

$g'(f(x))f'(x)=1$ ____(1)

$g''(f(x))=-\frac{f''(x)}{(f'(x))^3}$ ____(2)

$g'(3)=\frac{1}{23}$ $\quad g''(3)=\frac{-43}{5(23)^3}$ $\quad GE=\left[\left|g''(3)\right|\right]=0$

4. $a+b+c=13-3\sqrt{2}$ $\quad\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=13-3\sqrt{2}$

Produced $GE=\frac{(169-18)-1}{30}=\frac{150}{30}=5$

5. $x\left(1-x^2\right)^{-1}=x\left(1+x^2+x^4....\right)=x+x^3+x^5+...$

$\cos x\left(\frac{x}{1-x^2}\right)^{-1}=\cos\left(x+x^3\right)=1-\frac{\left(x+x^5\right)^2}{2!}+\frac{\left(x+x^3\right)^4}{4!}...$

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$$=1-\frac{x^2}{2}-x^4+\frac{x^4}{24}=1-\frac{x^2}{2}-\frac{23}{24}x^4$$

Also denominator $= x^2(1-\cos x)=x^2.2\sin^2\frac{x}{2}\approx x^2.2\left(\frac{x^2}{4}\right)\approx\frac{x^4}{2}$

$$\therefore \beta=\frac{\left(1-\frac{1}{2}x^2\right)-\left(1-\frac{x^2}{2}-\frac{23}{24}x^4\right)}{\frac{x^4}{2}}=\frac{23}{12}\Rightarrow 12\beta=23$$

Sum of digits = 5

6. $GE=A^{2024}-(A+2(I)).(A+2I)=0$

But $|A|=-1\neq 0\left(A^{-1}\ exists\right)$ $\quad |A+2I|=\begin{vmatrix}-2\beta & +2 & 0\\ 3 & 3 & -3\\ 4 & 1 & -1\end{vmatrix}=1(-1=-1\neq 0)$

$$\Rightarrow |A+2\beta I|=0 \quad \Rightarrow \begin{vmatrix}0 & 0 & 1\\ 3 & 1+2\beta & -3\\ 4 & 1 & -3+2\beta\end{vmatrix}=0\Rightarrow 3-4-8\beta=0\Rightarrow\beta=-\frac{1}{8}$$

$20\beta=-25\Rightarrow GE=3$

7. $(x-r)^2+y^2=r^2$ _____(1)

$\frac{x^2}{16}+\frac{y^2}{4}=1$_____(2)

Touch $\Rightarrow$ solving and $\Delta=0$ $\quad\Rightarrow r=\sqrt{3}$

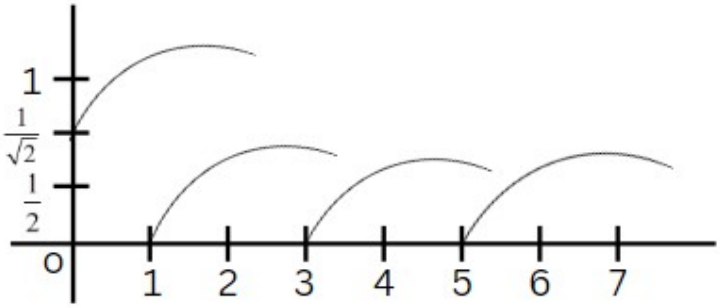
$\left(\because \frac{3}{4}x^2-2rx+4=0\Rightarrow 4r^2-12\Rightarrow r=\sqrt{3}\right)$

$$P=\left(\sqrt{3},0\right),Q=\left(-\sqrt{3},0\right),R=\left(\frac{4}{\sqrt{3}},\frac{2\sqrt{2}}{\sqrt{3}}\right),S=\left(-\frac{4}{\sqrt{3}},\frac{2\sqrt{2}}{\sqrt{3}}\right)$$

Area of trapezium P is $2S=\frac{1}{2}\left(2\sqrt{3}+\frac{8}{\sqrt{3}}\right)\times\frac{2\sqrt{2}}{\sqrt{3}}=\frac{14}{3}\sqrt{2}$ $\quad\lambda=3$

8. $4\pi\approx 12.56$ $\quad\Rightarrow x\in[0,12.56]$

One solution in each of $(1,3),(3,5),(5,7),(7,9),(9,11),(11,12.56)$



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No. of solutions = 6

9. $2\sin(B+C)+\frac{3}{2}(\cos(B+C)+\cos(B-C))=4$

$\Rightarrow 4\sin(B+C)+3\cos(B+C)+2\cos(B-C)=8$

$\therefore 4\sin(B+C)+3\cos(B+C)=5$ or $\cos(B-C)=1 \Rightarrow B=C$

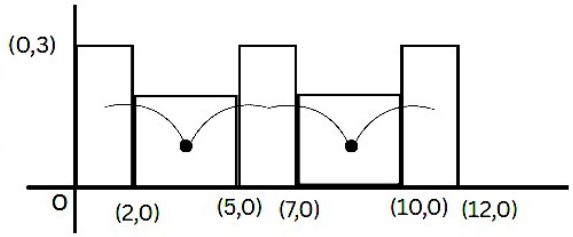
$\Rightarrow 4\sin 2B+3CD=B=5$

$\Rightarrow \frac{4(2t)}{1+t^2}+\frac{3(1-t^2)}{1+t^2}=5\Rightarrow t=\frac{1}{2}\Rightarrow \tan B=\frac{1}{2}\Rightarrow \sin B=\frac{1}{\sqrt{5}}$

$\sin A=\sin(B+C)=\sin 2B=\frac{2\times\frac{1}{2}}{1+\frac{1}{4}}=\frac{4}{5}$

10. Required area

$=4\left(\frac{1}{2}\right)+4(r)+7\frac{\pi}{2}=6+\frac{7\pi}{2}$



$a=6, b=7$

11. Solving 3 planes $(1,1,1)$ is part of intersection lies on $x-4y+\lambda z-1=0\Rightarrow \lambda=4$

$L_1:\frac{x-1}{1}=\frac{y-1}{-1}=\frac{z-1}{-1}$ $L_2:\frac{x-1}{0}=\frac{y-1}{1}=\frac{z-1}{1}$ $L_3:\frac{x}{0}=\frac{y}{0}=\frac{z}{1}$

$d(L_1,L_3)=\sqrt{2}, d(L_2,L_3)=1$

12. $a_{ij}=1 \quad \forall i=j$

$a_{ij}+a_{ji}=0 \quad$ when $i\neq j$

$\Rightarrow A-I$ is skew symmetric matrix

13. $J=\frac{1}{\sqrt{2}}\int_0^{\pi/4}\frac{e^{\sec x}.\sin(\sin x+\cos x)}{(1-\sin x)\cos x}dx$

Divides Nr & Dr with y^2x.

$J=\frac{1}{\sqrt{2}}\int_0^{\pi/4}e^{\sec x}.\sec x\tan x\left(\frac{1}{\sec x-\tan x}+\frac{\sec x(\sec x-\tan x)}{(\sec x-\tan x)^2}\right)dx$

$=\frac{1}{\sqrt{2}}\left(\frac{e^{\sec x}}{\sec x-\tan x}\right)_0^{\pi/4}=\frac{1}{\sqrt{2}}\left(\frac{e^{\sqrt{2}}}{\sqrt{2}-1}-e\right)=\frac{(1+\sqrt{2})e^{\sqrt{2}}-e}{\sqrt{2}}$

14. Conceptual

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15. A) $H=\dfrac{2\sqrt{6}}{\sqrt{3}+\sqrt{2}}\Rightarrow GE=3\times 2=6$

B) $\sum_{n=1}^{\infty}\dfrac{1}{2^n}\sum_{k=1}^{\infty}\dfrac{k}{2^k}(1)(2)=2$

C) $L=\dfrac{1}{18}\Rightarrow 72L=4$

D) Period = 2

16. $n_1=n(R_1)=6+4+3+3+(2\times 10)=36$

$n_2=n(R_2)=2\times 14=28$

Primes = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47}

$R_3=\{(1,2),(2,3),(2,1),(3,2),(1,3),(3,1),(1,1),(2,2),(3,3)\}$

$n_1=2^2.3^2 \qquad n_2=2^2.7^1$

$n_2+n_3=35=5^1.7^1$

$n_1+n_2+n_3=71 \quad n_3=7$

17. $y=mx-\dfrac{a}{4m},(1,0)$ lies or it $\Rightarrow a=4m^2$ ____(1)

But $(1,0)$ lies on directrix of hyperbola $\Rightarrow \dfrac{2}{e}=1\Rightarrow e=2$

$\Rightarrow 1+\dfrac{b^2}{4}=4 \qquad \Rightarrow b^2=12$

Trigonometry to hyperbola $y=mx\pm\sqrt{4m^2-b^2}$

$\Rightarrow \dfrac{a^2}{16n^2}=4n^2-b^2\Rightarrow\dfrac{16m^4}{16n^2}=3n^2-b^2\Rightarrow m^2=4m^2-b^2$

$\Rightarrow 3m^2=12\Rightarrow m^2=4\Rightarrow a=16$

18. (A),(B)

$f(x)=\sqrt{(x-1)^2+(0-1)^2}+\sqrt{(x-2)^2+(0-5)^2}$

$=PA+PB\geq\sqrt{37}$ occurs at $\alpha=\dfrac{7}{6}\Rightarrow 6\alpha=7$

(C) No. of $= 2^{42}\Rightarrow K=42\Rightarrow\dfrac{K}{6}=7$

(D) $\left(\sqrt{3}\sin x-\cos x\right)^2+\left(1+\sqrt{3}\sin x\right)(1+\cos x)=0$

$\Rightarrow(a-b)^2+ab=0\Rightarrow a^2-ab+b^2=0$

$\Rightarrow\left(a-\dfrac{b}{2}\right)^2+\dfrac{3}{4}b^2=0 \qquad \Rightarrow a=0\,\&\,b=0$ No solution

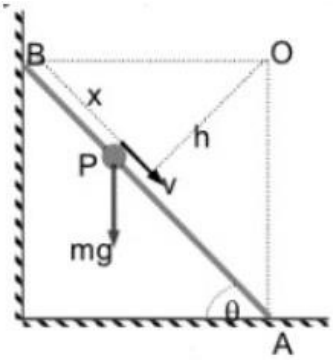
$\Rightarrow K=0\Rightarrow K+3=3$

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PHYSICS

19. For stable equilibrium $\frac{1}{h} > \frac{1}{r} - \frac{1}{R}$, here h is height of centre of mass of ring particle system.

20. Consider the bug at point P at any time t, moving with speed V along the stick,
sc $L = mvh = mvl\sin\theta\cos\theta \Rightarrow$ Angular momentum of bug about point O



$$\Rightarrow r = \frac{dL}{dt} = \left(ml\sin\theta\cos\theta\right)\frac{dv}{dt}$$

$$\Rightarrow mg\left(l-x\right)\cos\theta = \left(ml\sin\theta\cos\theta\right)\frac{dv}{dt}$$

$$\Rightarrow \frac{dv}{dt} = \frac{g}{\sin\theta}\left(1-\frac{x}{l}\right)$$

$$\Rightarrow \frac{dv}{dt} + \frac{g}{l\sin\theta}x = \frac{g}{\sin\theta} \Rightarrow \text{Equation of SHM with mean position at point A}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{l\sin\theta}{g}}$$

21. $$I_0 = \frac{k_0 A}{\cos\left(\frac{\pi}{6L}x\right)dx}$$

$$\Rightarrow T = I_0 = \frac{6I_0 L}{\pi k_0 A}\sin\left(\frac{\pi}{6L}x\right)$$

$$\Rightarrow T = 2I_0 \, at \, x = L$$

22. The area of the region where the CM can lie is the area of a circle with radius

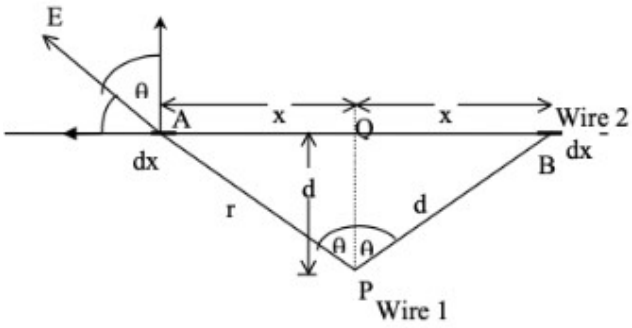
$$R_{CM,\max} = \frac{1}{2}m.$$

$$Area = \pi R_{CM,\max}^2 = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}m^2$$

23. $AP = r = \left(x^2 + d^2\right)^{1/2}$ Electric field at A due to wire 1

$$E = \frac{\lambda}{2\pi \in_0 r}$$

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Force on each elemental length dF =E.dq

$$dF = \left(\frac{\lambda}{2\pi \in_0 r}\right)(\lambda dx) = \frac{\lambda^2 dx}{2\pi \in_0 \left(x^2 + d^2\right)^{1/2}}$$

Net force on the elemental length $dF_{net} = 2dF\cos\theta$

$$= \frac{\lambda^2 dx\cos\theta}{\pi \in_0 \left(x^2 + d^2\right)^{1/2}} = \frac{\lambda^2 dx.d}{\pi \in_0 \left(d^2 + x^2\right)}$$

Hence total force required to hold the wire $\int dF_{net}$

$$F_{net} = \frac{\lambda^2 d}{\pi\in_0}\int_{-\infty}^{\infty}\frac{dx}{\left(d^2 + x^2\right)} = \frac{\lambda^2}{2\in_0} = \frac{\left(3\times10^{-6}\right)^2}{2\times8.86\times10^{-12}} = 0.51\,Newton.$$

As independent of separation between the wire. Hence required work done
$W = 0.51\times2 = 1\,Joule.$

24. The condition required for heating and melting the wire is that the amount of Joule heat liberated in the process must be larger than the amount of heat dissipated to the ambient: $I^2R > kS\left(T - T_{am}\right)$.

By hypothesis, the current required for melting the first wire must exceed 10 A.

Therefore, $I_1^2R_1 > k\left(\pi d_1 l\right)\left(T_{melt} - T_{am}\right)$

where l is the length of the wires, T_{melt} is the melting point of the wire material, and I_1 and R_1 are the current and resistance of the first wire.

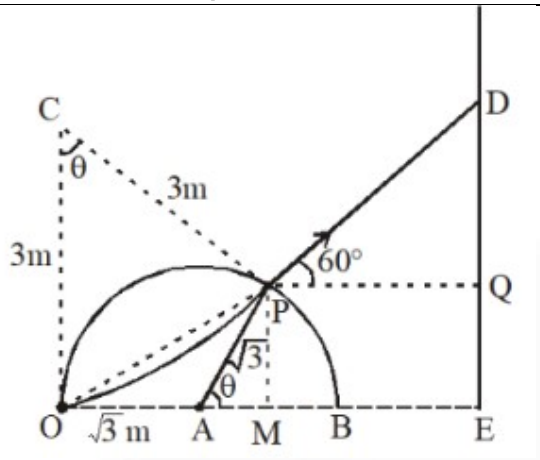
The resistance of the second wire is $R_2 = \frac{R_1}{16}$. Therefore, the current I_2 required for melting the second wire must satisfy the relation

$I_2^2R_2 \geq k.\pi d_2 l\left(T_{melt} - T_{am}\right)$

Finally, we obtain $I_2 \geq 8I_1 = 8.0A.$

25.

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Considering the upper half of the area

Radius of particle is circular motion $R=\frac{mv}{qB}=\frac{\left(2\times10^{-3}\right)(0.3)}{\left(1\times10^{-3}\right)\times0.2}=3m$

Radius of circular arc, $r=\sqrt{3}m$

From ΔCOP and ΔOAP

$$OP=2\times3\times\sin\frac{\theta}{2}=2\times\sqrt{3}\sin\left(\frac{180-\theta}{2}\right)$$

$$\Rightarrow \tan\left(\frac{\theta}{2}\right)=\frac{1}{\sqrt{3}}\Rightarrow\theta=60°$$

From, ΔAPM,

$$PM=\sqrt{3}\sin60°=\frac{3}{2}=\theta E$$

$$AM=\sqrt{3}\cos60°=\frac{3}{2}$$

From diagram,

$$PQ=ME=\sqrt{3}+\frac{\sqrt{3}}{2}=\frac{3\sqrt{3}}{2}$$

From ΔPQD,

$$\tan60°=\frac{DQ}{PQ}\Rightarrow DQ=\sqrt{3}\times\frac{3\sqrt{3}}{2}=\frac{9}{2}m$$

$$\therefore De=QE+DQ=\frac{3}{2}+\frac{9}{2}=6m$$

Similarlly we can get, from lower half

$\therefore$ Distance between the points where charges will strike after deflection $=6+6=12m=4x$

$\Rightarrow x=3$

26. Let frequency of rotating mirror be f .

$$\therefore \frac{2D}{V}\times2\pi f=\frac{\theta}{2}\Rightarrow D=\frac{25}{4}m$$

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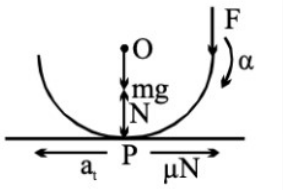
27. Torque due to applied force $\tau = Fr = I_P\alpha$

Where I_P is the moment of inertia about an axis passing through point of contact

$$I_0 = I_G + m(OG)^2 = mr^2 \Rightarrow I_G = mr^2 - m(OG)^2$$

$$I_P = I_G + m(GP)^2 = mr^2 - m(OG)^2 + m(r - OG)^2$$

$$= 2mr^2 - \frac{4mr^2}{\pi} \text{ where } OG = \frac{2r}{\pi} = \frac{2mr^2}{\pi}(\pi - 2)$$



Angular acceleration $\alpha = \frac{Fr\pi}{2mr^2(\pi-2)} = \frac{F\pi}{2mr^2(\pi-2)}$ in the clockwise direction

At $P, mg + F = N$, Friction force $f = \mu N + \mu(mg + F)$

For pure rolling, $a = \frac{\mu(mg+N)}{m} = \alpha(PG)$

$$\mu = \frac{m\alpha(PG)}{(mg+F)} = m\left[\frac{F\pi}{2mr(\pi-2)}\right]\frac{r(\pi-2)}{\pi(mg+F)} = \frac{F}{2(mg+F)}$$

28.



Small change in length 'dx' of rod when it is heated from T_0 to T_1 is

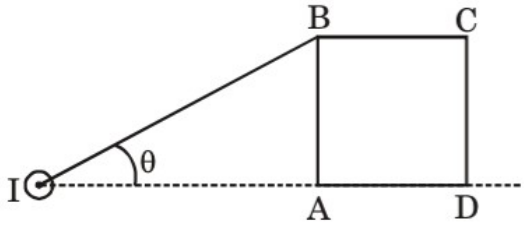
$$dl = \int_{T_0}^{T_1} dx(\alpha_0 + \alpha_1 x + \alpha_2 T)dT = \left[\alpha_0(T_1 - T_0) + \alpha_1 x(T_1 - T_0) + \frac{\alpha_2}{2}\left(T_1^2 - T_0^2\right)\right]dx$$

$\therefore$ Total change in length of the rod is

$$\int dl = \int_0^l \alpha_0(T_1 - T_0) + \alpha_1 x(T_1 - T_0) + \frac{\alpha_2}{2}\left(T_1^2 - T_0^2\right)dx$$

$$= \alpha_0(T_1 - T_2) + \frac{\alpha_1 l^2}{2}(T_1 - T_0) + \frac{\alpha_2 l}{2}\left(T_1^2 - T_0^2\right)$$

29.



$$\int_A^B \vec{B}.d\vec{l} = \frac{\mu_0 i}{2\pi}\theta$$

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So the line integral depends on θ only and not on path AB.

30. $R = R_0 A^{1/3}$ $A^{1/3} = 6$ $A_P = 216$

For daughter nucleus

$A_d = 212$, Also $\dfrac{212 - Z_d}{Z_d} = \dfrac{65}{41}$

$(41)(212) = 160 Z_d$ $Z_d = 82$ $Z_P = 84$

31. The capacitor in tweeter branch blocks the low frequency sound but passes the high frequency. The inductor in woofer branch does the opposite

$I_{ms1} = I_{rms2}$

$\Rightarrow \dfrac{1}{(\omega c)^2 + R^2} = \dfrac{1}{\left(\dfrac{1}{\omega c}\right)^2 + R^2}$ $\omega = \dfrac{1}{\sqrt{LC}} \Rightarrow f = \dfrac{1}{2\pi\sqrt{LC}}$

Voltage is same for both. In capacitor current leads voltage by $\cos^{-1} = \dfrac{R}{\sqrt{x_c^2 + R^2}} = \phi_1$

In inductor current leads voltage by $\cos^{-1} = \dfrac{R}{\sqrt{x_c^2 + R^2}} \Rightarrow \phi_2$

$\Rightarrow \phi_1 + \phi_2 = 90 \Rightarrow \cos(\phi_1 + \phi_2) = 0$

$\Rightarrow \tan\phi_1 \times \tan\phi_2 = 1$ $\dfrac{x_L}{R} \times \dfrac{x_C}{R} = 1 \Rightarrow L = CR^2$

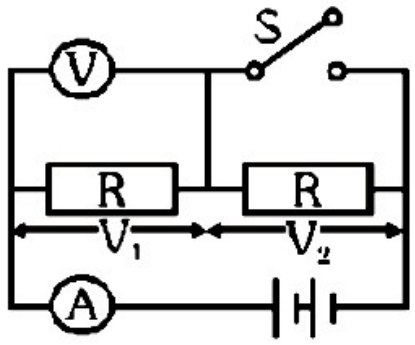
32. At contact point of man and plank, the relative motion is zero. As plank moves backward, man's foot also moves backwards, but friction on man acts forward. Hence work done by friction on man is negative. But with respect to plank, foot will be at rest, hence no work done by friction on man w.r.t plank.

33. Conceptual

34. Conceptual

35. Conceptual

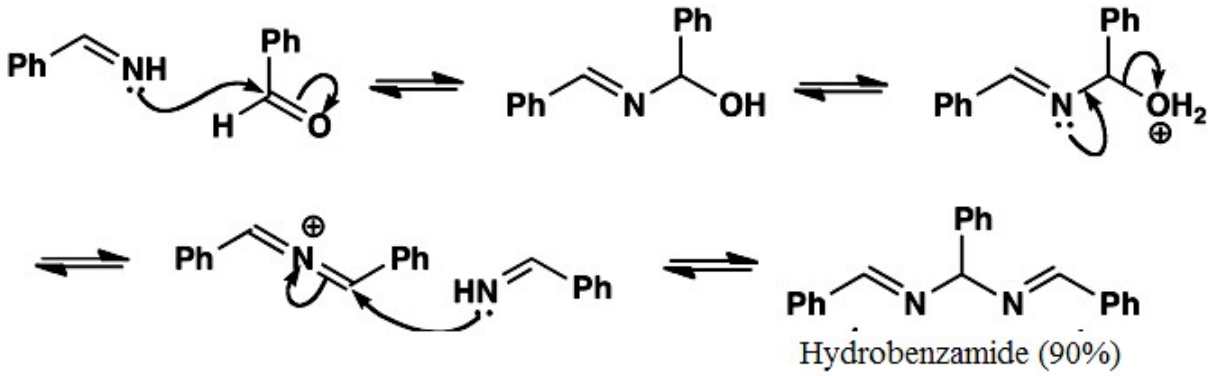
36. After closing the switch net resistance decreases therefore will be increases in the current. Initially $V = V_1 + V_2$ after closing the switch V_2 becomes zero hence $V = V_1$. After short circuiting current in the resistance becomes zero therefore power become zero.



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CHEMISTRY

37.



Hydrobenzamide (90%)

Imine formation follows the usual pathway but this imine is unstable, as are most primary imines, and it reacts with more benzaldehyde. This reaction starts normally enough but dehydration of the first intermediate produces a strange looking cation with two double bonds to the same nitrogen atom. Addition of another imine gives the final product. The benzene rings play no part in these reactions so we shall represent them as Ph, but they do stabilize the final product by conjugation with the imines.

38. Conceptual

39. pI of cysteine $= \dfrac{2.05+8}{2} = 5.025$

40. $Q = nC_{p_m}\Delta T; W_{out} = nR\Delta T$

$$\frac{Q}{W_{out}} = \frac{C_{p_m}}{R} = \frac{C_{v_m}+R}{R} = 1+\frac{C_{v_m}}{R} = \frac{7}{2}$$

$$\Rightarrow C_{v_m} = \frac{5R}{2}$$

41. $n_{NaOH} = n_{HCOOH} = n_{CH_3COOH} = 0.1$

OH^- is almost completely consumed by the acids

$$CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$$

eq^2 $\quad (0.1-x) - x$

$$HCOOH + OH^- \rightarrow HCOO^- + H_2O$$

$$(0.1-y) - y$$

$x + y \simeq 0.1$ -------(1)

Also $\left[H^+\right] = \dfrac{K_a\left[A^-\right]}{[HA]}$

$$\Rightarrow \left[H^+\right] = 2.5\times10^{-5}\left(\frac{0.1-x}{x}\right) = 10^{-4}\left(\frac{0.1-y}{x}\right)$$

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$$\Rightarrow\left(\frac{0.1-x}{x}\right)=4\left(\frac{x}{0.1-x}\right)\Rightarrow\left(\frac{x}{0.1-x}\right)^2=\frac{1}{4}$$

$$\Rightarrow x=\frac{0.1}{3}\Rightarrow(0.1-x)=\frac{0.2}{3}$$

$$\therefore\left[H^+\right]=10^{-4}\times\frac{[HCOOH]}{\left[HCOO^-\right]}$$

$$=10^{-4}\times\frac{0.1/3}{0.2/3}$$

$$=5\times10^{-5}\text{M.}$$

$$\frac{\left[In^-\right]}{[HIn]}=\frac{K_{in}}{\left[H^+\right]}=\frac{4\times10^{-10}}{5\times10^{-5}}=8\times10^{-6}.$$

42. $Fe_3O_4+4CO\longrightarrow3Fe+4CO_2$

To get 168 kg of iron, the amount of Fe_3O_4 required is 232kg.

To get 4.2 kg of iron, the amount of $Fe_3O_4=?$

$$\frac{4.2\times232}{168}\times\frac{100}{74}=7.83\text{kg}$$

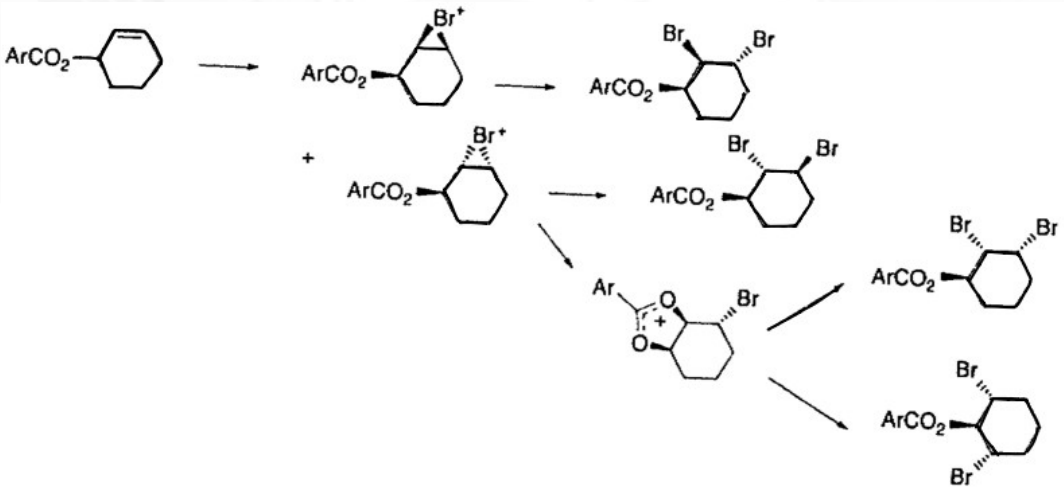
43. $CFSE=-2.4\Delta+2l$

$$=(-2.4\times25000)+\left(2\times15000\,cm^{-1}\right)$$

$$=30000\,cm=3\times10^4.$$

44. Square planar complex

45.



46. b) Due to $-I-$effect of Phenyl group

c) Due to $-I-$effect of $-OCH_3$ group less reactive than benzene in AES at meta position.

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47. $Fe^{2+} + Ce^{4+} \rightleftharpoons Ce^{3+} + Fe^{3+}$ $K = 10^{11}$ (From given E° values).

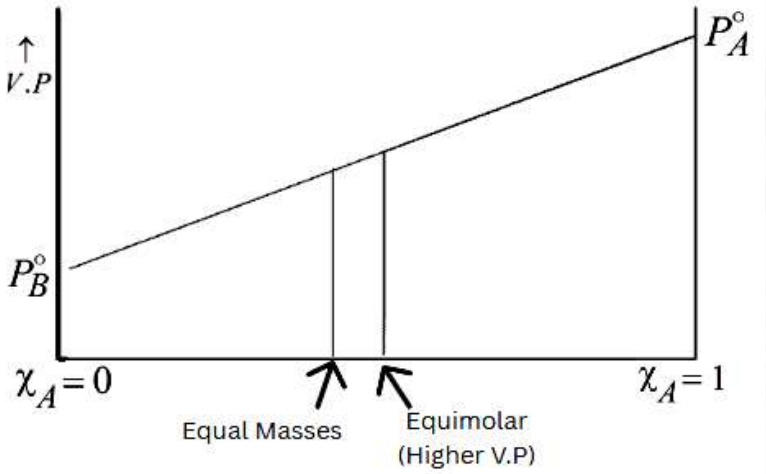
Ini moles: 11 1

Eqr moles: 10 - 1 1

$$10^{11} = \frac{1\times 1}{10.{}^{n}Ce^{4+}} \Rightarrow {}^{n}Ce^{4+} = 10^{-12}$$

$$\left[Ce^{4+}\right] = \frac{10^{-12}}{1.1} = 9.09\times 10^{-13}\text{M}.$$

48. If $MM_A > MM_B$, then $\chi_A < \frac{1}{2} < \chi_B$ for solution 2.



$\therefore P_A^\circ > P_B^\circ$

For Solution 2: $\chi_A = \frac{M/M_A}{\frac{M}{M_A}+\frac{M}{M_B}} = \frac{M_B}{M_A+M_B}; \chi_B = \frac{M_A}{M_A+M}$

$$Y_A = \frac{M_B P_A^\circ}{M_B P_A^\circ + M_A P_B^\circ}; Y_B = \frac{M_A P_B^\circ}{M_A P_B^\circ + M_B P_A^\circ}$$

$\because M_B < M_A$ and $P_A^\circ > P_B^\circ$,

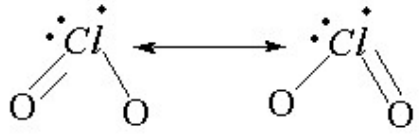
Y_A may be higher or lower than Y_B

49. Iron is in the +2 oxidation state. It is combined with S_2^{2-}. Sulphur exist as per sulphide ion $(S-S)^{2-}$. Persulphide ion is diamagnetic. Fe in +2 oxidation state contain 4 unpaired electrons. So magnetic moment is 4.9BM. On burning it forms Fe_2O_3 and SO_2.

50. Both OCl_2 and ClO_2, are angular $(V-\text{Shape})$ and can act as good oxidizing agents and bleaching agents. In OCl_2 the central atom is oxygen, involved in Sp^3 hybridization, carrying $2BP$ and $2LP$. Bond angle should be $109^\circ 28'$ but it is 111.2°. The increase is probably due to larger size of Cl atoms and partially due lack bonding from oxygen LP

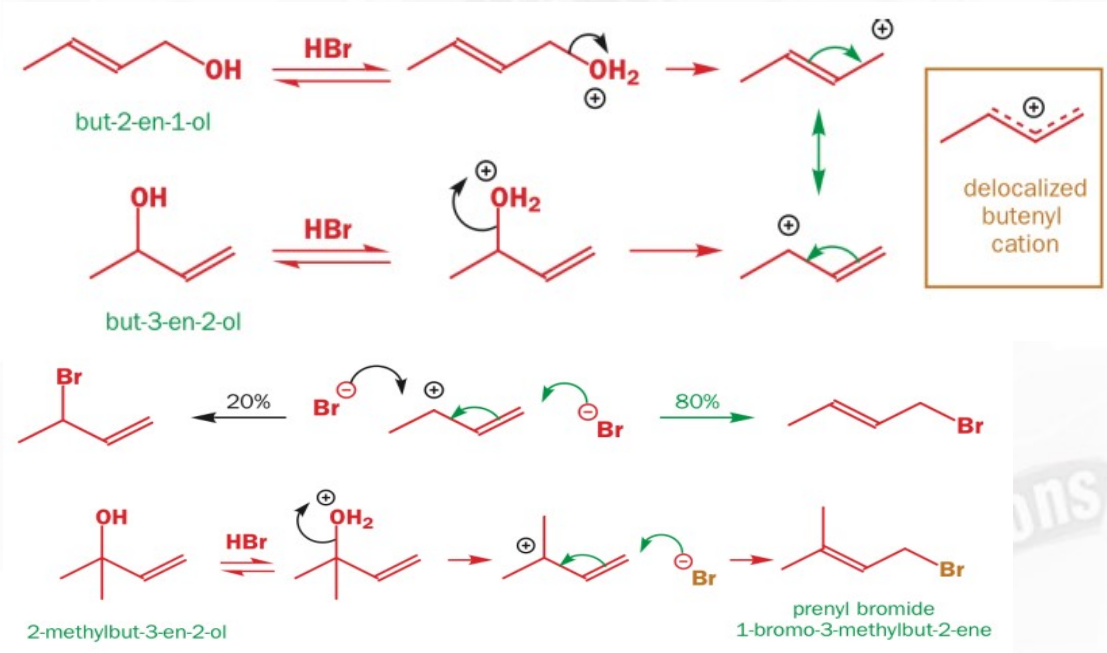
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in $p-$orbitals to vacant $d-$orbitals in chlorine atoms. In ClO_2 chlorine atom is involved in Sp^2 hybridization.



The unpaired electron also delocalizes participating in π bond. Thus the pair of electrons in π bond and the unpaired electron (3 electrons) delocalizes revolving round the three $O-Cl-O$ atoms. So the bond order in ClO_2 will be $3.5/2$ (Total number of bonds 1 double bond $\times 1$ single bond + 0.5 bond by one unpaired electron)=1.75, but in OCl_2 the bond order is $3.0/2$ (1 double bond + one single bond). So the $O-Cl$ bond length ClO_2 is 1.475A° and OCl_2 is 1.693Ω. Since the hybridization of Cl atom in ClO_2 is Sp^2. The bond length must be 120° but due to repulsion by lone pair, the bond angle decreases to 117.4°

51. In the reaction of allyl alcohols with HBr, the regioselectivity is determined by steric hinderence, Br^- attack is faster at less hindered side of allylic system.



52. Conceptual

53. e^- of X^{n+} is in $6s$ orbital. The only possible species for excitation as well as ionization are He^+ and Li^{2+} with $Li^{2+}, 6 \to 3$ will cause excitation in H from $1 \to 2$ with $He^+, 6 \to 2$ and $4 \to 2$ will cause excitation in H from $1 \to 3$ and $1 \to 2$ respectively. For H no ionization will take place and for Be^{3+} only ionization will take place.

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